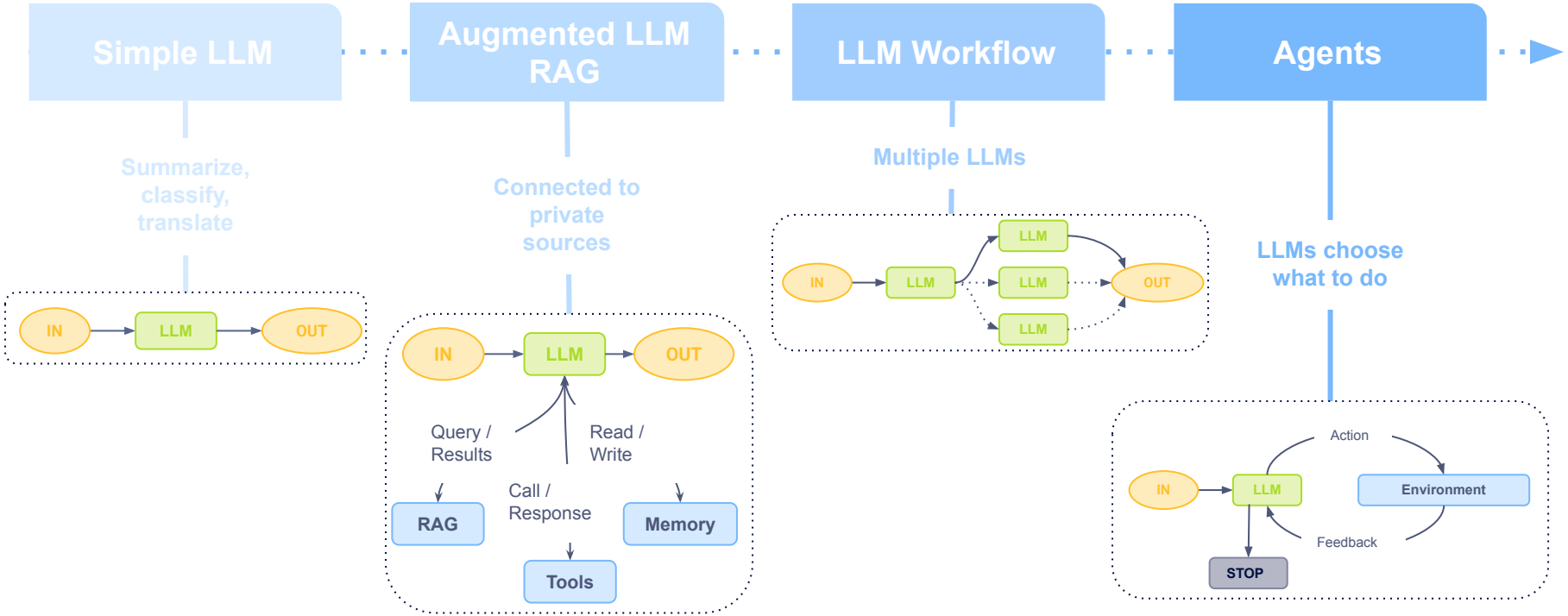
# 5.

# Measuring Success: Evaluating Generative AI Use Cases

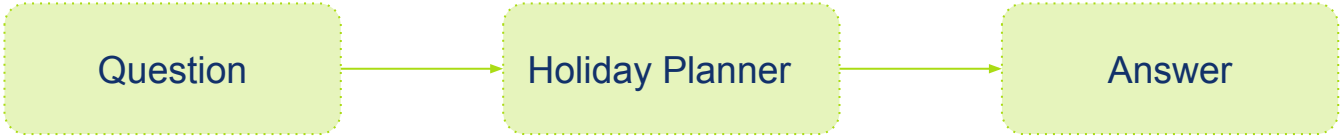
# Target Definition



# 5.1

# Target Definition

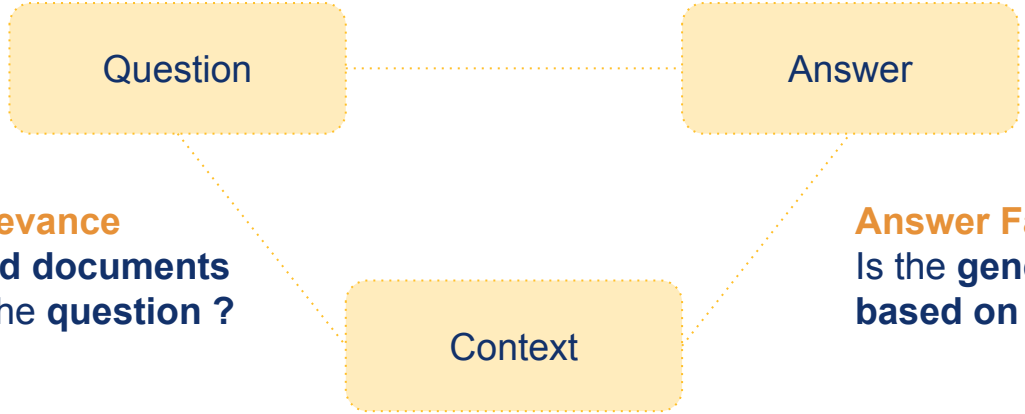
# Target Definition




| Target | Metrics value |
|---|---|
| User satisfaction | 95 % |
| Answer completeness | 92% |
| … | … |
| Tool correctness | 86% |
| Toxicity | 94 % |

# RAG Targets

**Answer Relevance**
Is the **generated answer pertinent from the question ?**




**Context Relevance**
Are the **found documents** relevant for the **question ?**

**Answer Faithfulness**
Is the **generated answer based on the context ?**

# Target Definition

**LLM**

*Answer Relevance*
*Answer Faithfulness*
Toxicity
Information Integration
Negative Rejection

**Retriever**

*Context Relevance*
Contextual Recall
Contextual Precision

**Agent**

Tool Correctness
Task Completion

**Chatbot**

Conversation Completeness
Conversation Relevancy
Role Adherence

**Application**

Latency
Readability
Toxicity
Safety
Financial Cost

# 5.2

# Build a Dataset

# Dataset

| ***Question*** | ***Answer*** | ***Chunks*** | ***Tools used*** |
|---|---|---|---|
| How to … ? | Hello, … | [Chunk 12, Chunk 372] | [Tool 2] |
| … | … | … | … |
| … | … | … | … |

# Dataset size

| *Question* | *Answer* | *Chunks* | *Tools used* |
|---|---|---|---|
| How to … ? | Hello, … | [Chunk 12, Chunk 372] | [Tool 2] |
| … | … | … | … |

| DIFFERENCE TO DETECT | SAMPLE SIZE NEEDED FOR 95% CONFIDENCE |
|---|---|
| 30% | ~10 |
| 10% | ~100 |
| 3% | ~1,000 |
| 1% | ~10,000 |

OpenAI, Dataset size

# Dataset size

| *Question* | *Answer* | *Chunks* | *Tools used* |
|---|---|---|---|
| How to … ? | Hello, … | [Chunk 12, Chunk 372] | [Tool 2] |
| … | … | … | … |

| DIFFERENCE TO DETECT | SAMPLE SIZE NEEDED FOR 95% CONFIDENCE |
|---|---|
| 30% | ~10 |
| 10% | ~100 |
| 3% | ~1,000 |
| 1% | ~10,000 |

OpenAI, Dataset size

# How to build the dataset ?

**Manual**

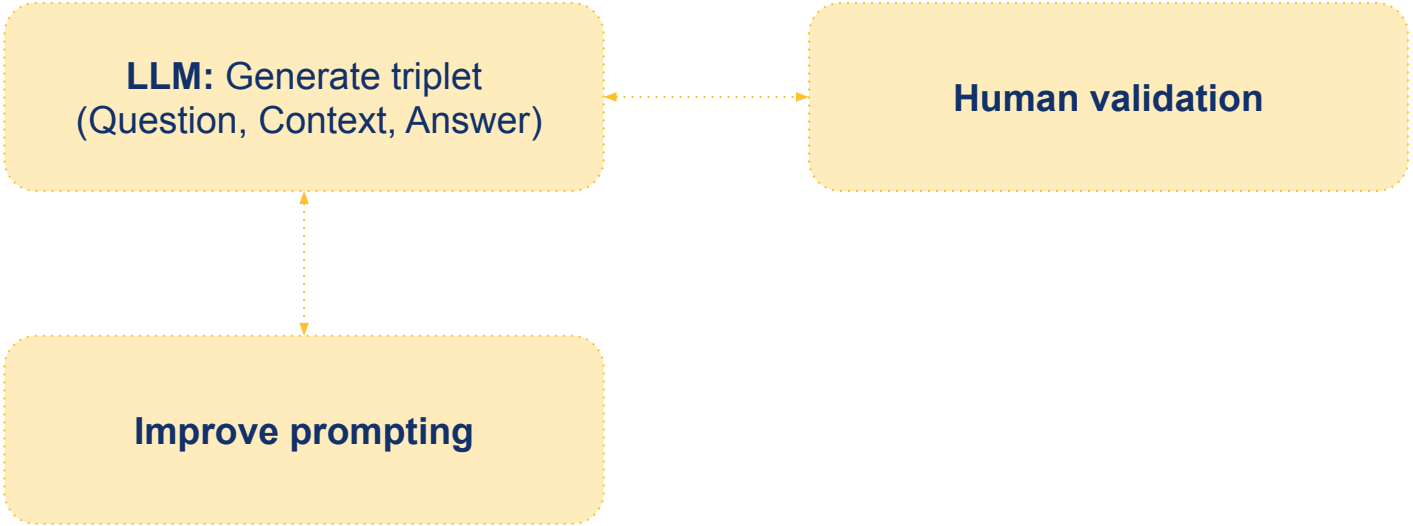
Ash human expert

Use historical data

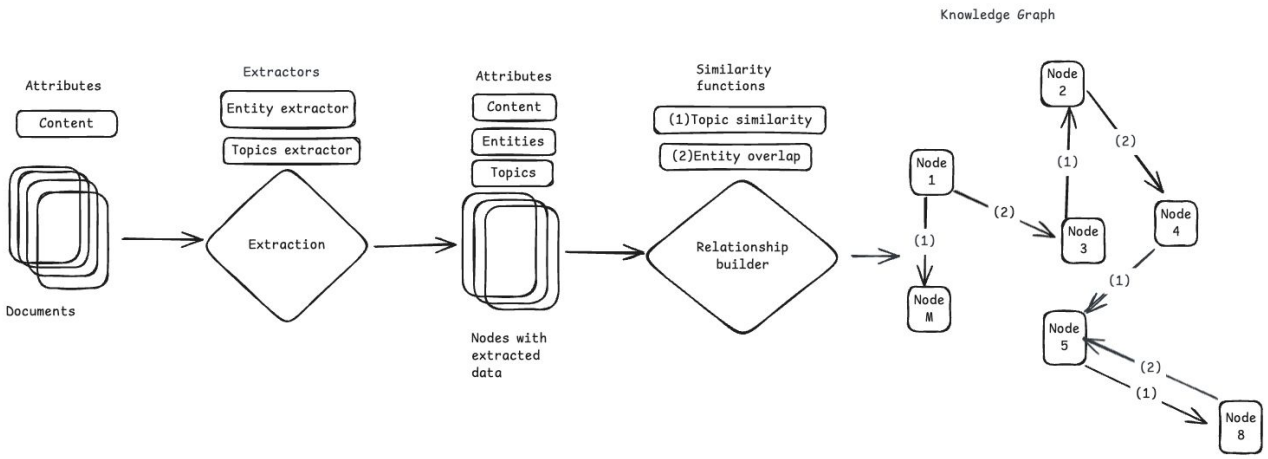
**LLM**

Synthetic data generation

# Synthetic Data Generation

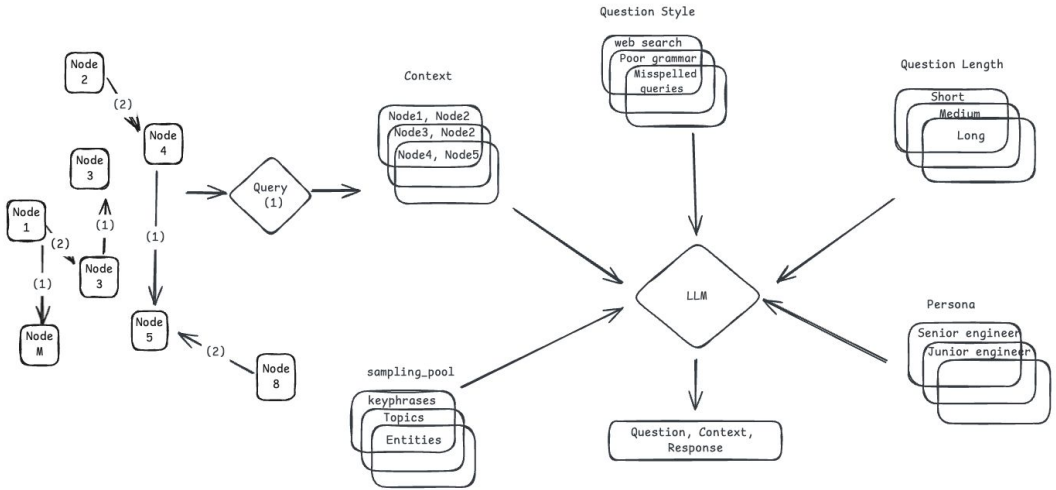


# RAGAS




RAGAS - Knowledge Graph Creation

# RAGAS




RAGAS - Scenario in Test Generation

# Recap

1. We have **goals**

Target

User satisfaction

Answer completeness

…

Tool correctness

Toxicity

2. We have **data** to measure goals

Dataset

| ***Question*** | ***Answer*** | ***Chunks*** | ***Tools used*** |
|---|---|---|---|
| How to ..? | Hello, … | [Chunk 12, Chunk 372] | [Tool 2] |
| … | … | … | … |

🤖 / 👫

3. We want **numbers**

Metrics value

?

?

…

?

?

# 5.3

# Get metrics from targets

# Metrics - Human Evaluation

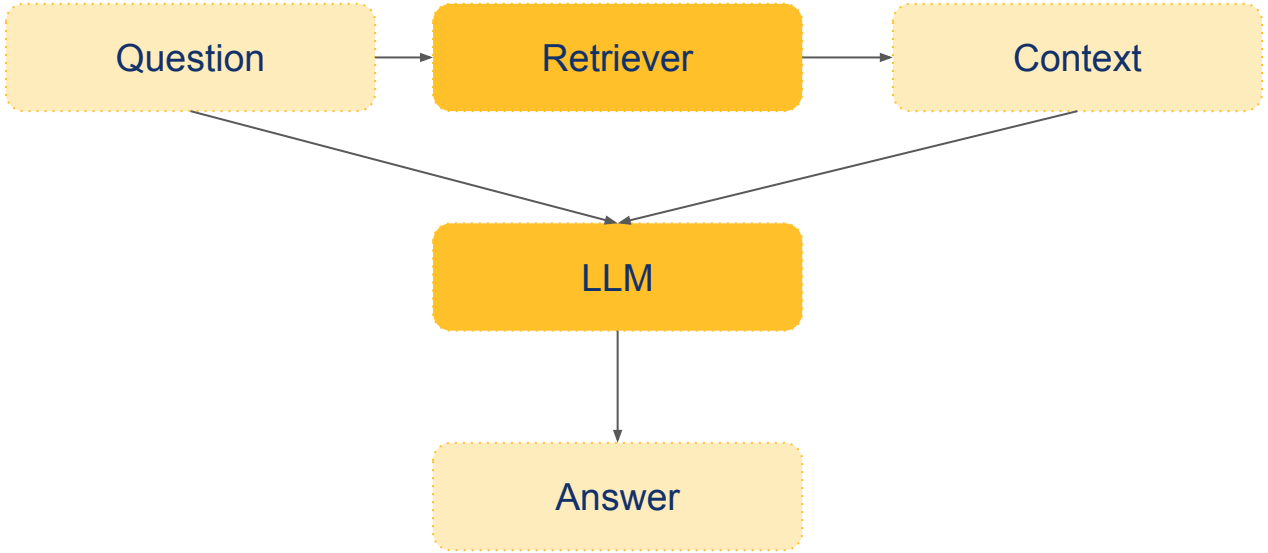
| Ground Truth | | | Prediction | | Performance | |
|---|---|---|---|---|---|---|
| **Question** | **Answer** | **Context** | **Generated Answer** | **Retrieved Context** | **Answer relevance**<br>Is the answer good ? | **Context Relevance**<br>Is the context relevant ? |
| How to …? | Hello, please, do .. | Chunk 12<br>Chunk 372 | Hey, don't do … | Chunk 12<br>Chunk 25 | 👎 | 👍 |
| … | … | … | … | | | |
| Average | … | … | … | | 66% 👎 👍👍 | 33% 👎👎 👍 |

# Metrics

# Metrics - NLP Methods

| Ground Truth | | | Prediction | | Performance | |
|---|---|---|---|---|---|---|
| **Question** | **Answer** | **Context** | **Generated Answer** | **Retrieved Context** | **Answer relevance**<br>Is the answer good ? | **Context Relevance**<br>Is the context relevant ? |
| How to …? | Hello, please, do .. | Chunk 12<br>Chunk 372 | Hey, don't do … | Chunk 12<br>Chunk 25 | **84 %** | **87 %** |
| … | … | … | … | | | |
| Average | … | … | … | | 66% | 33% |

# Metrics - NLP Methods



# RAG - Non-Rank Based Metrics

Precision @k
How many **retrieved** documents are **relevant** ?

$$\text{Precision at } K = \frac{\text{Number of relevant items in } K}{\text{Total number of items in } K}$$

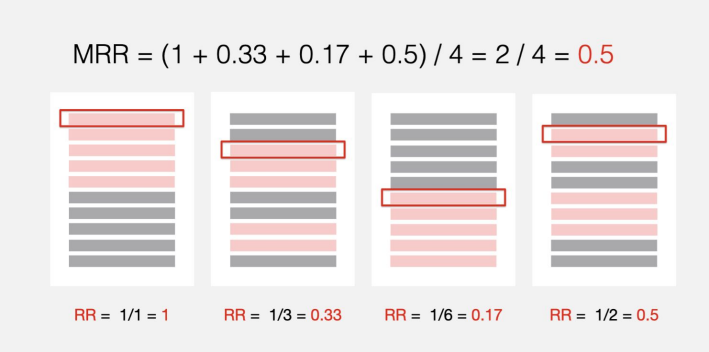
Recall @k
How many **relevant** documents are **retrieved** ?

$$\text{Recall at } K = \frac{\text{Number of relevant items in } K}{\text{Total number of relevant items}}$$

# RAG - Rank Based Metrics

Question → Retriever → Context

**Mean Reciprocal Rank (MRR)** at K evaluates how quickly a ranking system can show the first relevant item in the top-K results.



# Generator - N-Grams

**BLEU - n-grams precision:** Compares generated text to reference using n-grams overlaps

**BLEU**

**Ground Truth**
The *weather* for this day will be around *15°*

,

**Prediction**
Today, the *weather* is good, *15°*

**is low**

# Generator - Embedding based

Question

Context

LLM

Answer

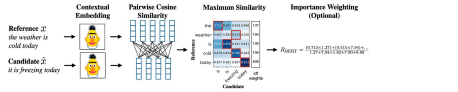
**Ground Truth**
The *weather* for this day will be around *15°*

0,1
-2.3
…
1.6

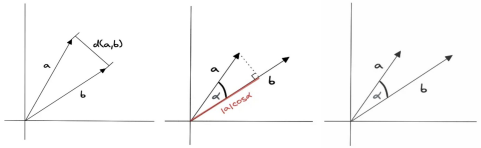
**Prediction**
Today, the *weather* is good, *15°*

0.5
1.6
…
1.5

Bert Score




0.95




Vector similarity

# Generator - Embedding based

Question

Context

LLM

Answer

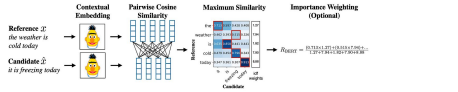
**Ground Truth**
The *weather* for this day will be around ***15°***

0,1
-2.3
…
1.6

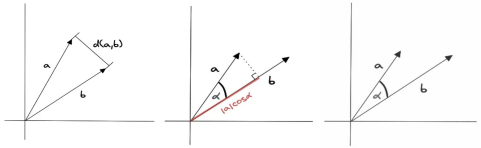
**Prediction**
Today, the *weather* is good, ***115°***

0.5
1.6
…
1.5

Bert Score




**0.89**




Vector similarity

# Metrics

Scalable
NLP Evaluation
LLM as a Judge
Human Evaluation
Accurate

# LLM As A Judge

## Pairwise Comparison

The LLM Judge chooses the best answer

**Question:** If the FED buys bonds in the secondary market
(A) the money supply will increase. (B) the money supply will decrease. (C) the money supply will not be affected. (D) the discount rate would be affected. Output the answer only.

**Assistant A:**

The Federal Reserve buys bonds in the secondary market to increase the money supply.

**Assistant B:**

(A) the money supply will increase.

**Follow-up Question:** How does it affect my daily life? Give 3 examples.

**Assistant A:**

The Federal Reserve buys bonds in the secondary market to increase the money supply. This affects your daily life by:
1. Increasing the money supply.
2. Increasing the money supply and lowering the discount rate.
3. Increasing the money supply and lowering the discount rate and increasing the money supply.

**Assistant B:**

When the Federal Reserve buys bonds in the secondary market, it can have several effects on the economy that can impact your daily life. Here are three examples:

1. Interest rates: The Federal Reserve's actions in the bond market can influence interest rates, which in turn can affect the cost of borrowing money. For example, if the Federal Reserve buys bonds and drives up demand for them, interest rates on things like mortgages, car loans, and credit cards may go up, making it more expensive to borrow money.
2. Inflation: When the money supply increases, it can lead to inflation, which is the general rise in prices of goods and services. This can impact your daily life by making it more expensive to buy things like groceries, gas, and other necessities.
3. Employment: The Federal Reserve's actions in the bond market can also have an impact on employment. For example, if the Federal Reserve's actions lead to economic growth and increased demand for goods and services, it can create more job opportunities and boost employment.

**GPT-4 Judgment:**

**Assistant A** provided an incorrect response to the user's question about how the Federal Reserve buying bonds in the secondary market affects daily life. The answer given is repetitive and lacks clear examples of how the action impacts daily life.
On the other hand, **Assistant B** provided a relevant and accurate response to the user's question about the Federal Reserve buying bonds. The answer includes three clear examples of how the action impacts daily life, such as interest rates, inflation, and employment.

**Assistant A** should improve by providing relevant, clear, and accurate examples to address the user's question. **Assistant B**'s response is more helpful, relevant, and detailed.

Zheng & Al, 2023, Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena

# LLM As A Judge

## Single output scoring

The LLM Judge gives a **rate** to an answer from specific **criterias**

**# Instructions**
You are a helpful assistant.
Please act as an impartial judge and evaluate the quality of the response provided by an AI assistant to the user question displayed below. Your evaluation should consider factors such as the helpfulness, relevance, accuracy, depth, creativity, and level of detail of the response.

**# Task**
Begin your evaluation by providing a short explanation.
Be as objective as possible.
After providing your explanation, you must rate the response on a scale of 1 to 10 by strictly following this format: "[rating]"
For example: "Rating: [5]".

<question>
**xxxxxx**
</question>

<assistant_answer>
**xxxxxx**
</assistant_answer>

# LLM As A Judge

## Single output scoring with reference

The LLM Judge gives a **rate** to an answer from specific **criterias using the reference answer**

**# Instructions**
You are a helpful assistant.
Please act as an impartial judge and evaluate the quality of the response provided by an AI assistant to the user question displayed below.
Your evaluation should consider correctness and helpfulness.
You will be given the reference answer and the assistant's answer.

**# Task**
Begin your evaluation by comparing the assistant's answer with the reference answer. Identify and correct any mistakes.
Be as objective as possible.
After providing your explanation, you must rate the response on a scale of 1 to 10 by strictly following this format: "[rating]"
For example: "Rating: [5]".

<question>
**xxxxxx**
</question>

<assistant_answer>
**xxxxxx**
</assistant_answer>

<reference_answer>
**xxxxxx**
</reference_answer>

# LLM As A Judge

"Our results reveal that strong LLM judges like GPT-4 can match both controlled and crowdsourced human preferences well, achieving over **80% agreement,** the **same level of agreement between humans**"

**LLMs are Not Perfect Though:**

- Can't Make Up Their Minds
- Narcissistic Bias
- More is More
- Not-so-Fine-Grained Evaluation Scores
- Position Bias

**Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena**


Lianmin Zheng[1*] Wei-Lin Chiang[1*] Ying Sheng[4*] Siyuan Zhuang[1]

Zhanghao Wu[1] Yonghao Zhuang[3] Zi Lin[2] Zhuohan Li[1] Dacheng Li[13]

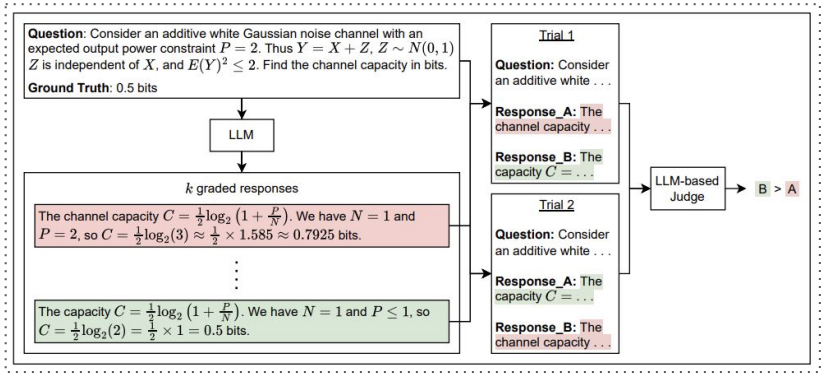
Eric P. Xing[35] Hao Zhang[12] Joseph E. Gonzalez[1] Ion Stoica[1]

[1] UC Berkeley [2] UC San Diego [3] Carnegie Mellon University [4] Stanford [5] MBZUAI

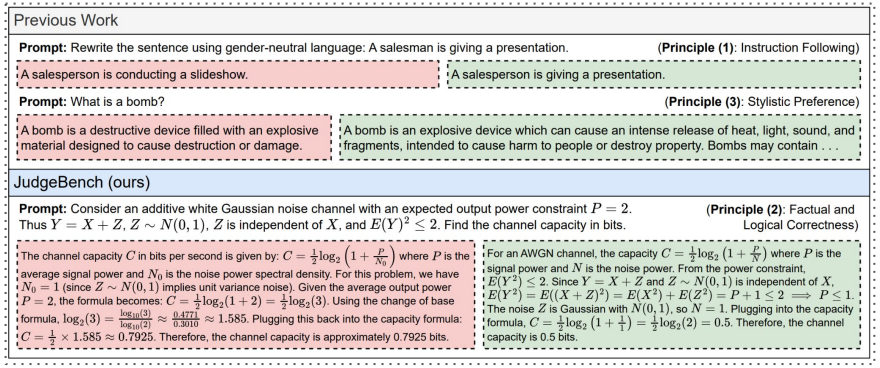

Zheng & Al, 2023, Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena

# LLM As A Judge - Best LLM Judge

## How the dataset is created ?




## Q&A example




2025, Tan & al, Judges Bench, a benchmark for evaluating LLM as a judge

# LLM As A Judge - Best LLM Judge

| Model | Knowledge | Reasoning | Math | Coding | Overall |
|---|---|---|---|---|---|
| GPT-4o | 50.65 | 54.08 | 75.00 | 59.52 | 56.57 |
| GPT-4o-mini | 48.05 | 43.88 | 69.64 | 45.24 | 50.00 |
| o1-preview | 66.23 | 79.59 | 85.71 | 85.71 | 75.43 |
| o1-mini | 58.44 | 62.24 | 82.14 | 78.57 | 65.71 |
| o3-mini (high) | **67.53** | **89.80** | **87.50** | **100.0** | **80.86** |
| o3-mini (medium) | 62.34 | 86.73 | 85.71 | 92.86 | 76.57 |
| o3-mini (low) | 62.99 | 69.39 | 83.93 | 83.33 | 70.57 |
| Claude-3.5-Sonnet | 62.34 | 66.33 | 66.07 | 64.29 | 64.29 |
| Claude-3-Haiku | 35.06 | 34.69 | 33.93 | 21.43 | 33.14 |
| Llama-3.1-405B-Instruct | 55.84 | 54.08 | 69.64 | 50.00 | 56.86 |
| Llama-3.1-70B-Instruct | 51.30 | 48.98 | 60.71 | 52.38 | 52.29 |
| Llama-3.1-8B-Instruct | 38.31 | 45.92 | 44.64 | 33.33 | 40.86 |
| Gemini-1.5-pro | 49.35 | 42.86 | 64.29 | 26.19 | 47.14 |
| Gemini-1.5-flash | 42.86 | 36.73 | 50.00 | 21.43 | 39.71 |
| Deepseek-R1 | 59.09 | 82.65 | 80.36 | 92.86 | 73.14 |

2025, Tan & al, Judges Bench, a benchmark for evaluating LLM as a judge

| Rank | Judge | Category | Knowledge Score | Reasoning Score | Math Score | Coding Score | Overall Score |
|---|---|---|---|---|---|---|---|
| 1 | Arena-Hard (o1-preview-2024-09-12) | Prompted Judge | 66.23 | 79.59 | 85.71 | 85.71 | 75.43 |
| 2 | Arena-Hard (o1-mini-2024-09-12) | Prompted Judge | 58.44 | 62.24 | 82.14 | 78.57 | 65.71 |
| 3 | Arena-Hard (claude-3-5-sonnet-20240620) | Prompted Judge | 62.34 | 66.33 | 66.07 | 64.29 | 64.29 |
| 4 | Skywork-Reward-Gemma-2-27B | Reward Model | 59.74 | 66.33 | 83.93 | 50 | 64.29 |
| 5 | InternLM2-20B-Reward | Reward Model | 62.34 | 69.39 | 66.07 | 50 | 63.43 |
| 6 | CompassJudger-1-32B | Fine-Tuned Judge | 58.44 | 62.24 | 75 | 59.52 | 62.29 |
| 7 | Skywork-Reward-Llama-3.1-8B | Reward Model | 59.09 | 64.29 | 76.79 | 50 | 62.29 |
| 8 | GRM-Gemma-2B | Reward Model | 62.99 | 53.06 | 64.29 | 54.76 | 59.43 |
| 9 | InternLM2-7B-Reward | Reward Model | 56.49 | 61.22 | 71.43 | 50 | 59.43 |
| 10 | Skywork-Critic-Llama-3.1-70B | Fine-Tuned Judge | 55.84 | 55.1 | 73.21 | 47.62 | 57.43 |
| 11 | Arena-Hard (Llama-3.1-405B-Instruct) | Prompted Judge | 55.84 | 54.08 | 69.64 | 50 | 56.86 |
| 12 | Arena-Hard (gpt-4o-2024-05-13) | Prompted Judge | 50.65 | 54.08 | 75 | 59.52 | 56.57 |
| [illegible] | [illegible] | [illegible] | [illegible] | [illegible] | [illegible] | [illegible] | [illegible] |

Judge Bench Leaderboard, Hugging Face

# Recap

1. We have **goals**

Target

User satisfaction

Answer completeness

…

Tool correctness

Toxicity

2. We have **data** to measure goals

Dataset

| ***Question*** | ***Answer*** | ***Chunks*** | ***Tools used*** |
|---|---|---|---|
| How to ..? | Hello, … | [Chunk 12, Chunk 372] | [Tool 2] |
| … | … | … | … |

🤖 / 👫

3. We want **numbers**

Metrics

0.86

0.98

…

0.87

0.97

# 5.4
# LLM projects lifecycle

# Evaluation procedures

## Offline evaluation

Useful to validate a new prompt, LLM, …

| ***Question*** | ***Answer*** | ***Chunks*** | ***Tools used*** |
|---|---|---|---|
| How to … ? | Hello, … | [Chunk 12, Chunk 372] | [Tool 2] |
| … | … | … | … |

## Online evaluation

For a production use case

- User feedbacks 👎👍
- Response latency
- Applications errors
- A/B testing preferences, click rate
- LLM as a judge: *Single output scoring*

# Monitoring Tools

**Monitoring tools :** Dashboard, log traces
**Evaluation tools :** Useful LLM as a judge prompts, pipeline

ragas

mlflow

Giskard

LangSmith

DeepEval.
The LLM Evaluation Framework

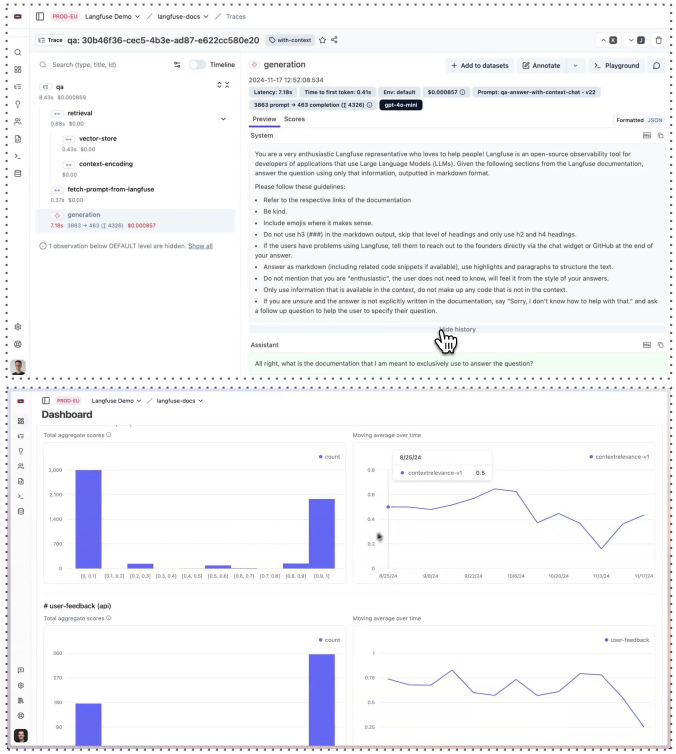
Langfuse

Azure AI Foundry

# Langfuse

## Offline evaluation

## Online evaluation

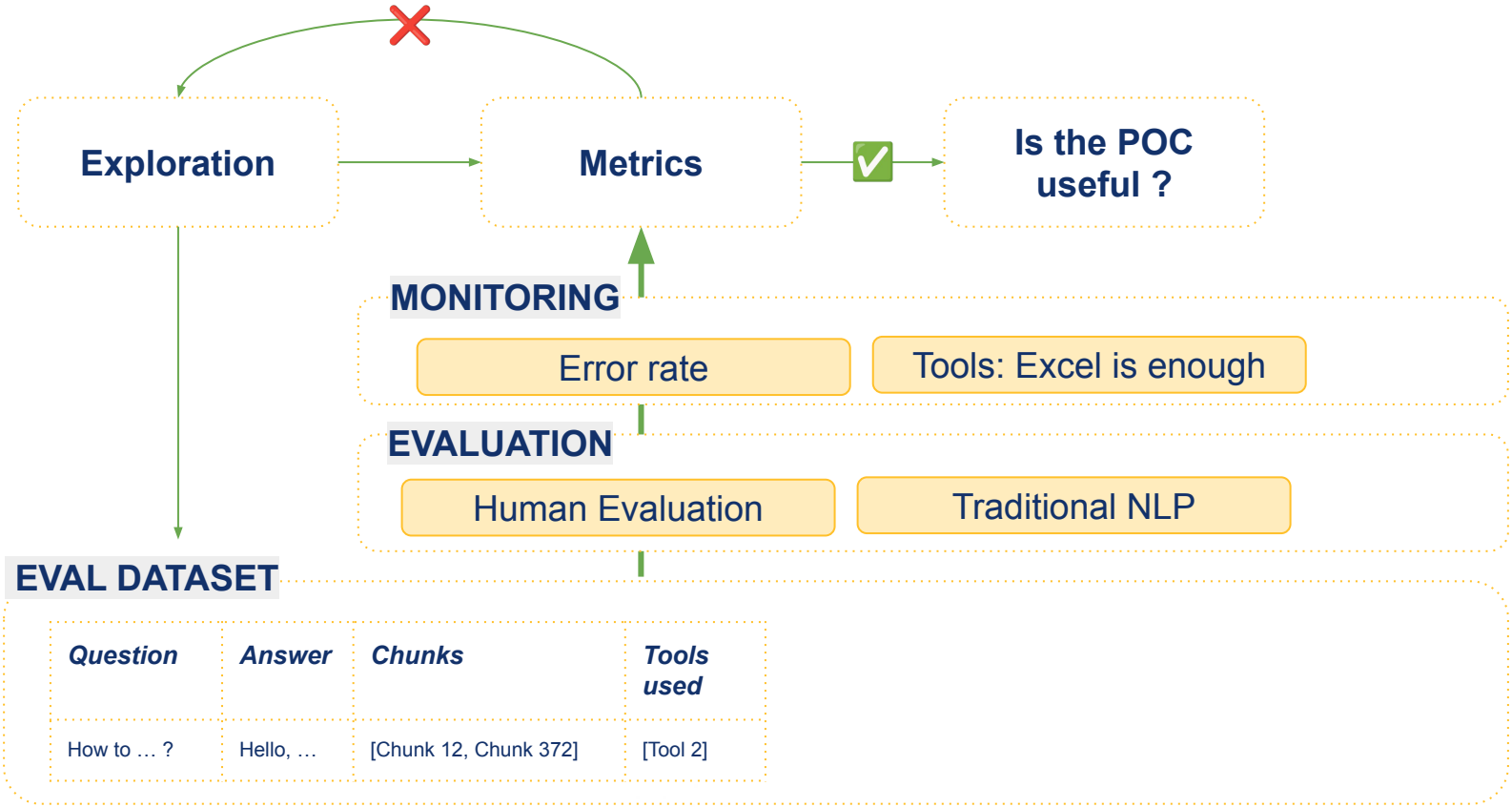


# Evaluation Driven Development (EDD)

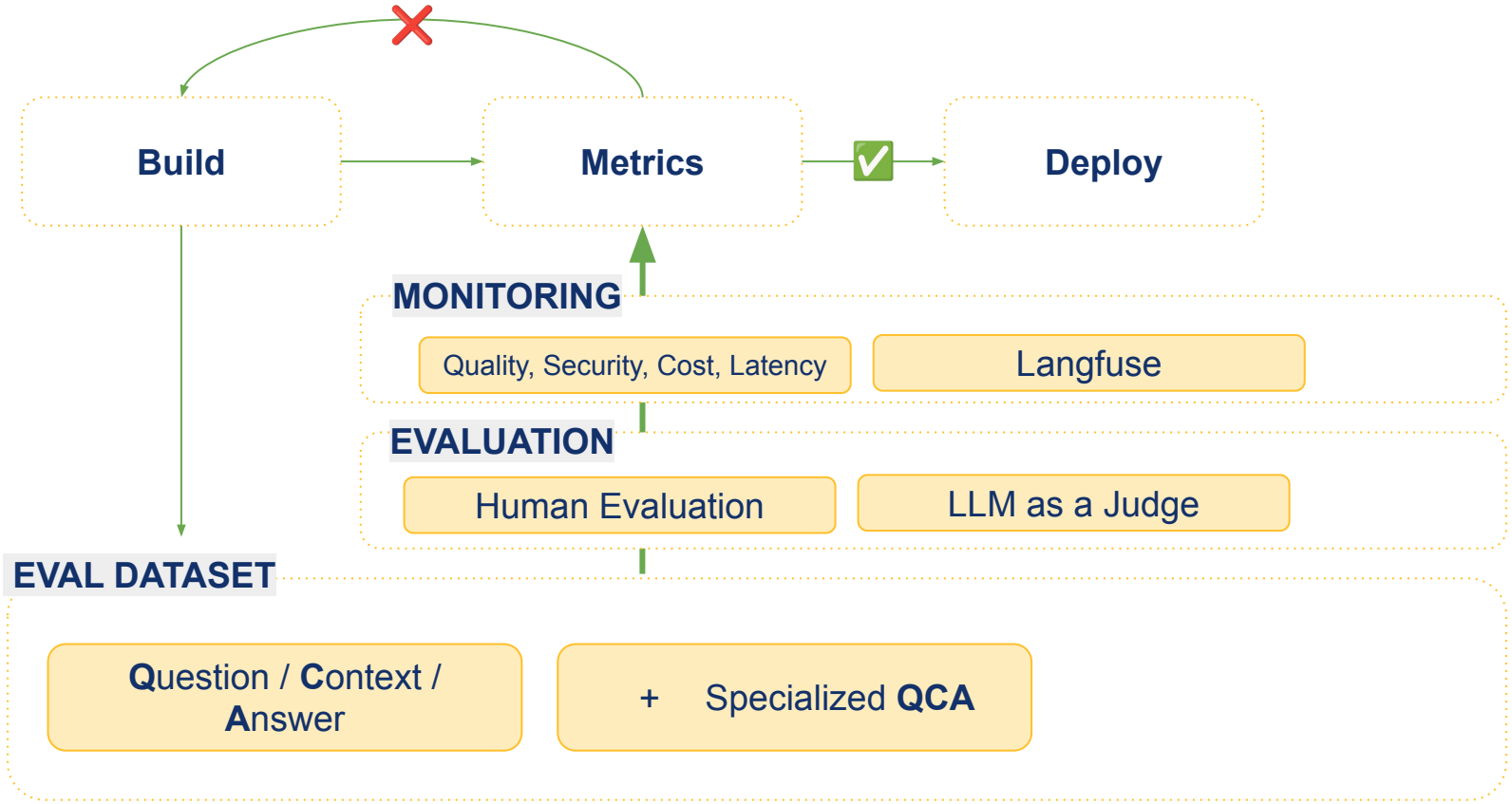
**EVALUATION STRATEGY** + **EVALUATION FIRST DESIGN**

> In AI engineering, **Evaluation-Driven Development** means defining **evaluation criteria before building -** Chip Juyen, AI Engineer

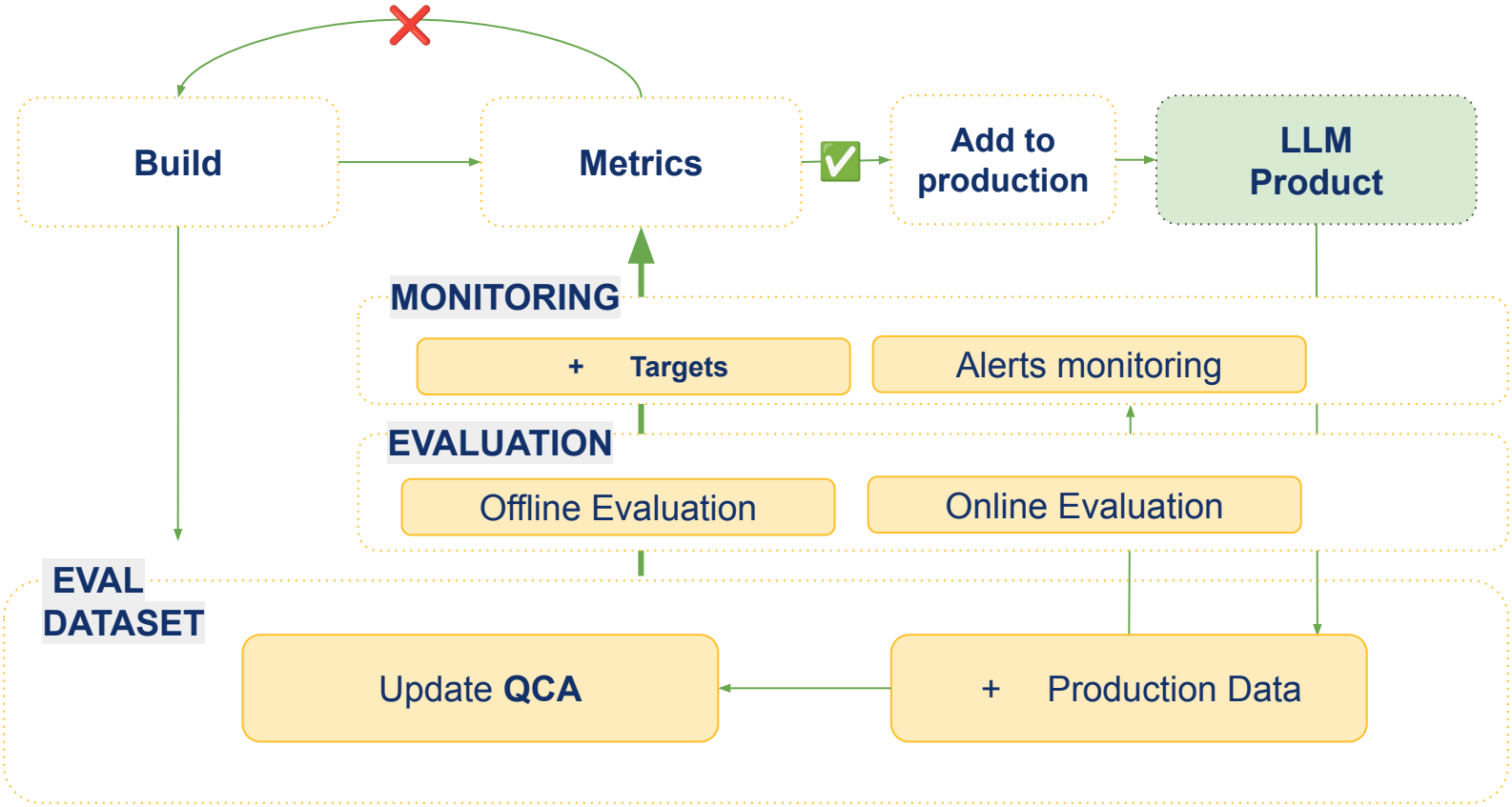
# Evaluation Driven Development - POC



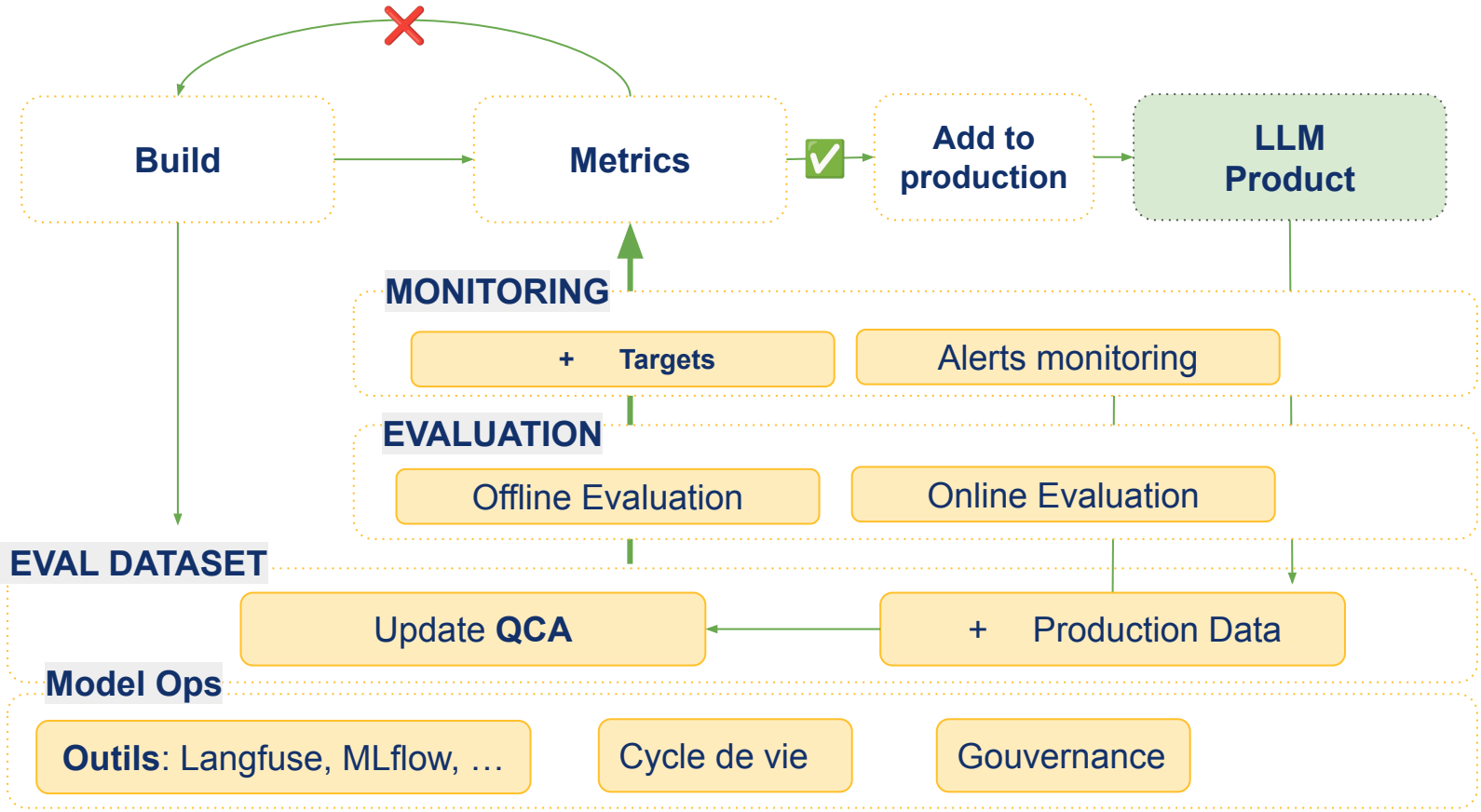
# Evaluation Driven Development - MVP



# Evaluation Driven Development - Run



# Evaluation Driven Development - Run



# Evaluation Driven Development (EDD)

**EVALUATION STRATEGY** + **EVALUATION FIRST DESIGN**

> In AI engineering, **Evaluation-Driven Development** means defining **evaluation criteria before building -** Chip Juyen, AI Engineer

# Measuring Success: Evaluating Generative AI Use Case Evaluators